Varianza y Prueba de Hipotesis

Varianza: La varianza se define como la dispersión de los datos alrededor de la media:

$$Var(X) = E(X - EX)^2$$

$x_1, x_2, \ldots, x_n$

$$E[x] = \frac{\sum_{i=1}^{n} x_i}{n} \qquad Var(x) = \frac{\sum (x_i - \mu)^2}{n} \; ; \quad \mu = \frac{\sum x_i}{n}$$

Covarianza: Sean X y Y variables aleatorias. La covarianza entre X e Y se define como:

$$\mathrm{Cov}(X, Y) = E((X - EX)(Y - EY))$$

$$Cov(x, x) = E((x - Ex)(x - Ex)) = E((X - Ex)^2)$$

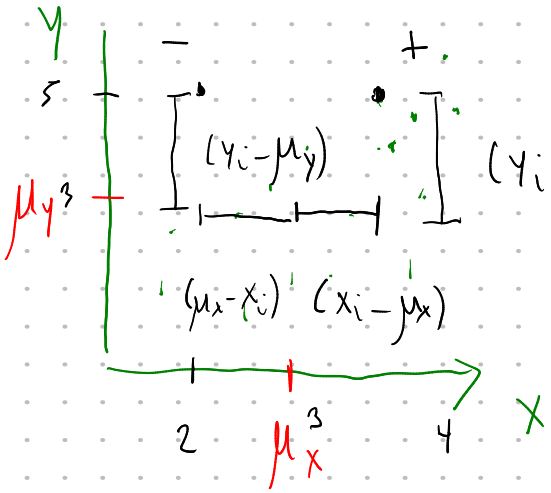
$$= Var(x)$$

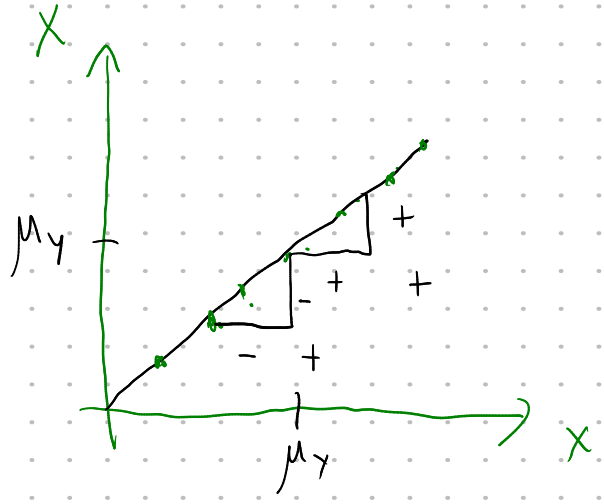
$$= E((X_i - \mu_x)(Y_i - \mu_y))$$

$$Cov(x, y) = \frac{\sum_{i=1}^{n} (x_i - \mu_x)(Y_i - \mu_y)}{n} \; ; \quad \mu_x = \frac{\sum x_i}{n}$$

$$\mu_y = \frac{\sum y_i}{n}$$

$$Cov(X,Y) = \frac{\sum_{i=1}^{n} (x_i - \mu_x)(y_i - \mu_y)}{n}$$

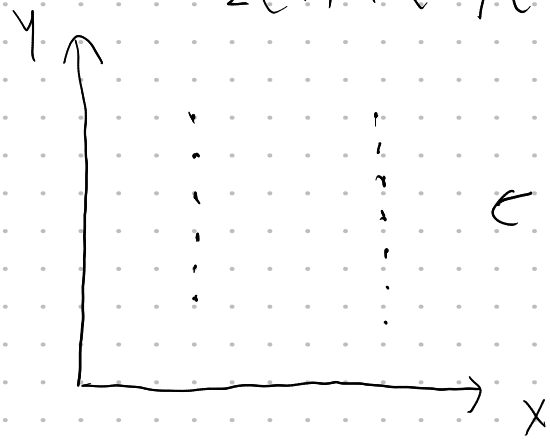




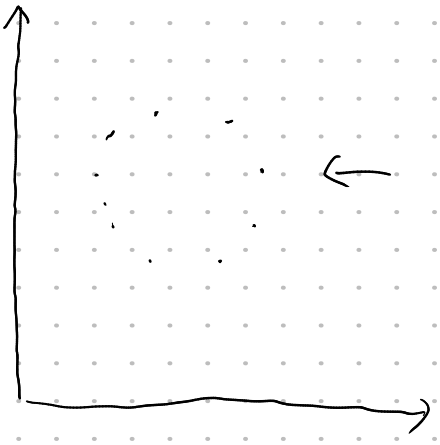
$$(Y_1 - \mu_Y)(X_1 - \mu_X) + (X_2 - \mu_X)(Y_2 - \mu_Y)$$

$$(5-3)(3-2) + (5-3)(4-3)$$

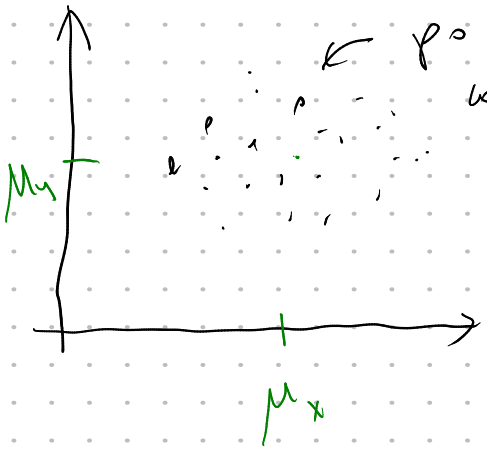
$$2(1) + (2)(-1) = 0$$



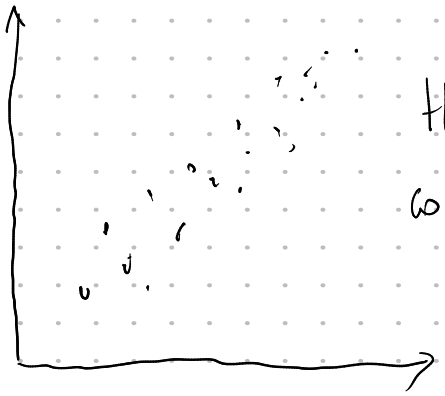
← 0 Cov



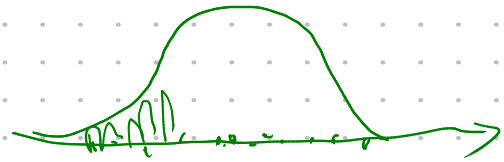
← 0 Cov

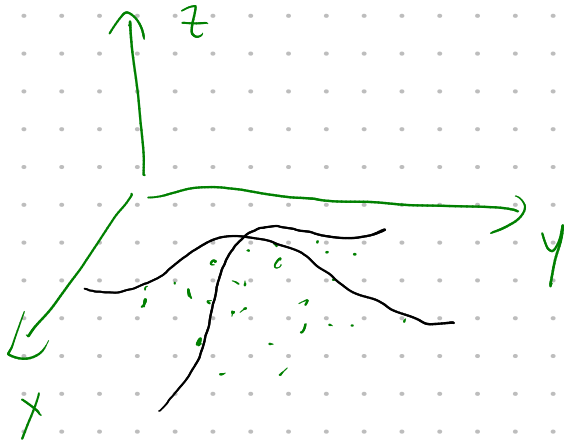


← poca covarianza



Harta covarianza





$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

No hay correlación $\quad cov(x,y)=0 \rightarrow f(x,y) = f(x)f(y)$

$$f(x,y) = \frac{1}{\sqrt{2\pi\sigma_x^2}} \frac{1}{\sqrt{2\pi\sigma_y^2}} e^{-\frac{(x-\mu_x)^2}{2\sigma_x^2}} e^{-\frac{(y-\mu_y)^2}{2\sigma_y^2}}$$

$$e^{-\frac{1}{2\sigma_x^2}(x-\mu_x)^2 - \frac{1}{2\sigma_x^2}(y-\mu_y)^2}$$

Si $cov(x,y) \neq 0 \qquad f(x,y) \neq f(x)f(y)$

Matriz de covarianza

$$\Sigma = \begin{bmatrix} var(x) & cov(x,y) \\ cov(y,x) & var(y) \end{bmatrix} \qquad \vec{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$$

The probability density function for **`multivariate_normal`** is

$$f(x) = \frac{1}{\sqrt{(2\pi)^k \det \Sigma}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right),$$

$$\vec{x}_i = \begin{bmatrix} x_i \\ y_i \end{bmatrix} \qquad (\vec{x} - \vec{\mu})_i = \begin{bmatrix} x_i - \mu_x \\ x_i - \mu_y \end{bmatrix} \qquad (x-y)_i^T = [x_i - \mu_x \quad y_i - \mu_y]$$

$$\langle \psi | H | \psi \rangle \qquad (x-\mu)^T \Sigma^{-1} (\vec{x} - \vec{\mu}) = \sum [x_i - \mu_x \quad y_i - \mu_y] \begin{bmatrix} var(x) & cov(x,y) \\ cov(y,x) & var(y) \end{bmatrix} \begin{bmatrix} x_i - \mu_x \\ y_i - \mu_y \end{bmatrix}$$

vector fila conjugado $\qquad$ vector columna

$$(1,2)(2,2)(2,1) = (1,1)$$

Teoria del error

Medición.

$X_1, X_2, \dots, X_n$

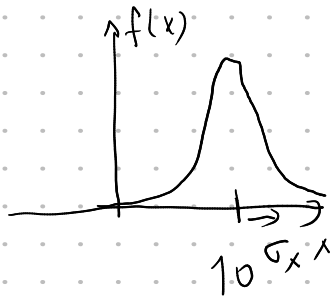
Reporto los datos

$\bar{X} \pm \sigma_x$

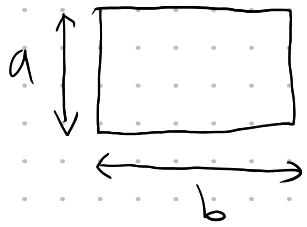
$\bar{X} \pm \Delta X$

σ_x: Desviación Est

σ_x^2: Varianza



$$\sigma_x \approx \Delta X$$



$\{a_1, \dots, a_n\}, \{b_1, \dots, b_n\}$

$\mu_a \pm \sigma_a \quad , \quad \mu_b \pm \sigma_b$

$\bar{a} \pm \Delta a \quad , \quad \bar{b} \pm \Delta b$

$$c = a + b = \frac{P}{2} \qquad \mu_c = \frac{\sum a_i + b_i}{n} = \frac{\sum a_i}{n} + \frac{\sum b_i}{n}$$

$$\mu_c = \mu_a + \mu_b$$

$$\bar{c} = \bar{a} + \bar{b}$$

$$\Delta c \approx \Delta a + \Delta b$$

$$\Delta c \approx \sqrt{(\Delta a)^2 + (\Delta b)^2}$$

$$\sigma_c^2 = E[(c - E[x])^2] = \frac{1}{n}\sum_{i=1}^{n}(c_i - \mu_c)^2 \qquad \mu_c = \frac{\sum c_i}{n}$$

$$c_i = a_i + b_i \qquad \mu_c = \mu_a + \mu_b$$

$$\sigma_c^2 = \frac{1}{n}\sum(a_i + b_i - \mu_a - \mu_b)^2 = \frac{1}{n}\sum((a_i - \mu_a) + (b_i - \mu_b))^2$$

$$= \frac{1}{n}\sum[\underbrace{(a_i - \mu_a)^2}_{(1)} + \underbrace{(b_i - \mu_b)^2}_{(2)} + \underbrace{2(a_i - \mu_a)(b_i - \mu_b)}_{(3)}]$$

$$\sigma_c^2 = \sigma_a^2 + \sigma_b^2 + 2\,\mathrm{Cov}(a,b) \leftrightarrow \mathrm{Var}(A,B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A,B)$$

$$\sigma_c^2 \approx \sigma_a^2 + \sigma_b^2 \leftarrow$$ Para la teoria del error.

$$\boxed{\Delta c^2 \approx (\Delta a)^2 + (\Delta b)^2}$$

$$c = a - b \Rightarrow \sigma_c^2 = \sigma_a^2 + \sigma_b^2$$

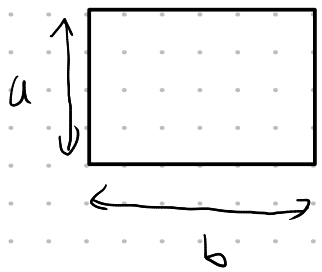
para la suma y resta $\{ (\Delta c)^2 \approx (\Delta a)^2 + (\Delta b)^2$

Regla del producto y del cociente.

$$\frac{(\Delta c)^2}{c^2} = \frac{(\Delta a)^2}{a^2} + \frac{(\Delta b)^2}{b^2}$$

$$\Delta c = c\sqrt{\left(\frac{\Delta a}{a}\right)^2 + \left(\frac{\Delta b}{b}\right)^2}$$

$\{a_1, \dots, a_n\}, \{b_1, \dots, b_n\}$



Area: $c = ab$

$$\mu_c = \frac{1}{n}\sum (x_i y_i)$$

$$\mu_a \mu_b = \left(\frac{1}{n}\sum x_i\right)\left(\frac{1}{n}\sum y_i\right)$$

$$= \frac{1}{n^2}\left(\sum x_i\right)\left(\sum y_i\right) = \frac{1}{n^2}(x_1 + \dots + x_n)(y_1 + \dots + y_n)$$

$$= \frac{1}{n^2}\sum x_{ij} = \frac{1}{n^2}\left[\sum_{i=1}^{n} x_i y_i + \sum_{i \neq j} x_i y_j\right]$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} c_i$$

$$\mu_c = \mu_a \mu_b$$

$$Var(c) = Var(a,b) = Var(a)Var(b) + var(a)\mu_b^2 + var(b)\mu_a^2$$

$$\sigma_c^2 = \sigma_a^2 \sigma_b^2 + \sigma_a^2 \mu_b^2 + \sigma_b^2 \mu_a^2$$

$$(\Delta c)^2 = (\Delta a)^2 (\Delta b)^2 + (\Delta a)^2 \bar{b}^2 + (\Delta b)^2 \bar{a}^2$$

$$(\Delta a)^2 (\Delta b)^2 << (\Delta a)^2 \bar{b} + (\Delta b)^2 \bar{a}^2$$

$$\left[\sigma_c^2 \simeq \sigma_a^2 \mu_b^2 + \sigma_b^2 \mu_a^2 \right] \frac{1}{\mu_a^2 \mu_b^2}$$

$$\frac{\sigma_c^2}{\mu_a^2 \mu_b^2} \simeq \frac{\sigma_a^2}{\mu_a^2} + \frac{\sigma_b^2}{\mu_b^2}$$

$$\frac{\sigma_c^2}{\bar{c}^2} = \frac{\sigma_a^2}{\bar{a}^2} + \frac{\sigma_b^2}{\bar{b}^2}$$

Formula para el producto y el cociente.

$$x = x_0 + v_0 t$$

Datos: $\bar{x}_0, \sigma_0, \bar{v}, \sigma_v, \bar{t}, \sigma_t$

$$\left[\frac{\Delta(v_0 t)}{(v_0 t)}\right]^2 = \frac{\sigma_v^2}{\bar{v}^2} + \frac{\sigma_t^2}{\bar{t}^2}$$

$$\Delta(v_0 t) = v_0 t \sqrt{\left(\frac{\sigma_v}{\bar{v}}\right)^2 + \left(\frac{\sigma_t}{\bar{t}}\right)^2}$$

$$(\Delta x)^2 = (\Delta x_0)^2 + \Delta(v_0 t)^2$$

$$\sigma_x = \sqrt{(\sigma_{x0})^2 + (v_0 t)^2\left[\left(\frac{\sigma_v}{\bar{v}}\right)^2 + \left(\frac{\sigma_t}{\bar{t}}\right)^2\right]}$$

$$Var(x+y) = Var(x) + Var(y) + Cov(x, y)$$

$$X_1, X_2, X_3, \ldots, X_n$$

$$Var\left(\sum X_i\right) = \sum_{i=1}^{n} Var(X_i) + \sum_{i \neq j} Cov(X_i, X_j)$$

$$Var(x+y+z) = Var(x) + Var(y+z) + Cov(x, y+z)$$

$$\sigma_{\bar{x}}^2 = \sigma_{x_1}^2 + \sigma_{x_2}^2 + \cdots + \sigma_{x_n}^2$$

p-value: Considere una $\bar{X}$ el promedio de una muestra de n datos que viene de una distribución de μ desconocida y σ conocida. Se quiere probar la hipotesis de que $\bar{X}$ tiene viene de una distribución con promedio μ_0. Sea $\Phi(x)$ la función acumulativa de la función normal $\mathcal{N}(0,1)$. El p-value se define como,

$$2 * \Phi(\left|\frac{\bar{X} - \mu_o}{\sigma/\sqrt{n}}\right|) \tag{2}$$

Example

In a certain chemical process, it is very important that a particular solution that is to be used as a reactant have a pH of exactly 8.20. A method for determining pH that is available for solutions of this type is known to give measurements that are normally distributed with a mean equal to the actual pH and with a standard deviation of .02. Suppose 10 independent measurements yielded the following pH values:

8.18	8.17
8.16	8.15
8.17	8.21
8.22	8.16
8.19	8.18

(a) What conclusion can be drawn at the $\alpha = .10$ level of significance?
(b) What about at the $\alpha = .05$ level of significance?